

Project Documentation Format

1. Introduction

- **Project Title:** Automated Network Request Management in ServiceNow

Team Members: list the people on the project and their roles. A suggested breakdown for this project:

Role	Responsibility
Project Lead / Solution Owner	Owns overall solution design and stakeholder sign-off
ServiceNow Configurator / Developer	Builds the catalog item, Flow Designer flows, business rules, and ACLs
Business Analyst	Gathers requirements from network operations and defines approval/routing rules
Network Operations SME	Validates request types, CMDB data, and provisioning steps
QA / ServiceNow Tester	Builds ATF test suites, executes UAT, validates approval and notification scenarios

2. Project Overview

- **Purpose:** To replace the manual, email- and spreadsheet-driven process for requesting network access and changes (VPN access, firewall rules, new circuits, VLANs) with a single, automated ServiceNow workflow that validates, routes, approves, and provisions requests with minimal human intervention.
 - A guided Service Catalog item with a dynamic form that adapts to the type of network request.
 - Automated validation of each request against the CMDB and network policy rules before it reaches an approver.
 - Rules-based approval routing (manager, security, or network owner) with automatic reminders and out-of-office rerouting.
 - Automated provisioning via IntegrationHub to firewalls, VPN concentrators, and SD-WAN controllers once a request is approved.
 - Real-time status tracking, notifications, and a post-fulfillment CSAT survey for the requester.
 - Reporting dashboards for SLA compliance, request volume, and recurring request types.

3. ServiceNow Components & Configuration

This project is built entirely on the ServiceNow platform using out-of-the-box and low-code capabilities — there is no separate frontend, backend, or custom API layer to maintain. The table below lists the components used and what each one is responsible for.

Component	Used For
Service Catalog Item	Guided, dynamic form the requester fills in to request network access or a change
Catalog Client Scripts & UI Policies	Show/hide and validate fields depending on the request type selected
Flow Designer Flows	Orchestrate submit → validate → approve → provision → notify → close, with no custom code
Business Rules & Script Includes	Server-side validation logic, e.g. checking the CMDB before a request can proceed
IntegrationHub Spokes	Pre-built or custom actions that talk to firewalls, VPN concentrators, and SD-WAN controllers
CMDB (Configuration Management Database)	Source of truth for the network devices/CIs a request affects

Approvals & Notifications	Route requests to the right approver and keep the requester informed automatically
Performance Analytics / Reporting	Dashboards for SLA compliance, request volume, and recurring request types

4. Setup Instructions

- **Prerequisites:** A ServiceNow Personal Developer Instance (PDI) or sandbox instance; a ServiceNow account with admin rights on that instance; the Flow Designer and IntegrationHub plugins activated; access credentials (or a sandbox/mock endpoint) for the network orchestration tool(s) being integrated.
- **Configuration steps:** A step-by-step guide to bring the environment up:
 - Provision or request access to a ServiceNow PDI/sandbox instance.
 - Import the update set containing the catalog item, custom tables, flows, and business rules.
 - Create the Service Catalog category and item(s) for network requests, and attach the catalog client scripts and UI policies.
 - Activate and configure the Flow Designer flows for validation, approval routing, and provisioning.
 - Configure IntegrationHub connection & credential aliases (and a MID Server, if required) for the target network systems.
 - Assign roles and configure ACLs so requesters, approvers, network engineers, and administrators see only what they should.

5. Configuration Structure

Instead of a code folder structure, the project is organized into the following ServiceNow configuration artifacts, all contained within a single scoped application:

- Service Catalog category, catalog item, and its variable set (the request form).
- Custom Network Change Request table, extending Task, with fields for source/destination, ports, environment, and the related CI.
- Flow Designer flows and subflows: Submit & Validate, Approval Workflow, Provision Network Change, Notify & Close.
- Business Rules and Script Includes used for CMDB validation and routing-rule lookups.
- Notification records and email templates for each stage of the request.
- Reporting: a Performance Analytics dashboard (or a set of reports) built on top of the request table.

6. Running the Process End-to-End

There is no build/start command as with a coded application; the process runs directly on the ServiceNow instance:

1. Publish the Service Catalog item so it is visible to the requester's role/group.
2. Activate the associated Flow Designer flows (Submit & Validate, Approval Workflow, Provisioning, Notify & Close).
3. Use "Try It" on the catalog item (or the Service Portal preview) to submit a test request end-to-end.
4. Monitor execution under Flow Designer > Flow Executions, and inspect the generated RITM/Task records.

7. Access Control & Approvals

- **Sign-in:** Requesters and approvers sign in via the organization's existing SSO, so no separate ServiceNow credentials are required.

- **Record-level security:** Access Control Lists (ACLs) restrict who can view, create, or update Network Change Request records — for example, only the assigned approver can act on a pending approval, and only members of the Network Operations group can update provisioning tasks.
- **Approvals:** Approval routing rules determine who needs to sign off (manager, security, or network owner) based on the request type and business unit, with automatic reminders and out-of-office rerouting.

8. Screens & Forms Used

All screens are standard ServiceNow forms and portal widgets — no custom UI was built. Key screens to capture with screenshots once available:

- The network request catalog item, showing the dynamic form fields for a selected request type.
- The requester's "My Requests" view, showing status, RITM number, and history.
- The approver's Approvals inbox / mobile approval card.
- The Network Engineer's task list for provisioning and escalations.
- The reporting dashboard showing SLA compliance and request volume.

9. Testing

- **Automated testing:** ServiceNow's Automated Test Framework (ATF) is used to script and run regression tests for the catalog form, business rules, and Flow Designer flows on every update-set promotion.
- **Integration testing:** A sandbox/mock endpoint for the network orchestration tool is used to test the provisioning flow without touching production network devices.
- **User acceptance testing (UAT):** Business stakeholders (network operations, security, a sample of requesters) walk through representative scenarios — happy path, rejected approval, provisioning failure and escalation — before each release.
- **Performance testing:** Volume testing simulates peak submission periods (e.g., end-of-quarter project rollouts) to confirm the approval and provisioning flows do not bottleneck.

10. Demo

Video Link: <https://drive.google.com/file/d/1AbY7UbVsRuhjXsVG8cu3e8ApAqaI2LUk/view?usp=sharing>

11. Known Issues

Issue	Impact	Status
Approval delegation does not follow nested out-of-office chains (A delegates to B, B is also OOO)	Approval can stall until someone manually reassigns it	Open
Provisioning integration to one VPN vendor's API silently times out under rate limiting instead of raising a retryable error	Requires manual detection and re-submission by the network engineer	Open
Dynamic catalog form occasionally does not reset dependent fields when the requester changes request type mid-form	Minor UX confusion; does not affect submitted data integrity	Open
CMDB validation step can time out for very large VLAN/subnet lookups	Adds a few seconds of delay for a small subset of requests	Monitoring

12. Future Enhancements

- AI-assisted request classification that auto-suggests the correct catalog item and pre-fills fields from a free-text description.
- ChatOps integration (Slack/Microsoft Teams virtual agent) so requesters and approvers can act without opening the portal.
- Predictive SLA-breach alerts that flag at-risk requests before they miss their deadline.

- Self-service diagnostics (e.g., automated connectivity tests) offered before a ticket is even created, to deflect requests that don't need a change.
- Expanded vendor coverage for the provisioning integration (additional firewall, SD-WAN, and cloud network vendors).